

The equivalence principle and spacetime curvature

We are now in a position to use the experience gained in deriving a relativistic formulation of electromagnetism (together with some flashes of inspiration from Einstein!) to begin our formulation of a relativistic theory of gravity, namely general relativity.

7.1 Newtonian gravity

In our development of electromagnetism, we began by considering the electromagnetic 3-force on a charged particle. Let us therefore start our discussion of gravity by considering the description of the gravitational force in the classical, non-relativistic, theory of Newton. In the Newtonian theory, the gravitational force $\vec{f}$ on a (test) particle of *gravitational* mass m_G at some position is

$$\vec{f} = m_G \vec{g} = -m_G \vec{\nabla} \Phi,$$

where $\vec{g}$ is the gravitational field derived from the *gravitational potential* Φ at that position. In turn, the gravitational potential is determined by Poisson's equation:

$$\boxed{\vec{\nabla}^2 \Phi = 4\pi G \rho,} \tag{7.1}$$

where ρ is the *gravitational matter density* and G is Newton's gravitational constant. This is the field equation of Newtonian gravity.

It is clear from (7.1) that Newtonian gravity is *not* consistent with special relativity. There is no explicit time dependence, which means that the potential Φ (and hence the gravitational force on a particle) responds *instantaneously* to a disturbance in the matter density ρ ; this violates the special-relativistic requirement that signals cannot propagate faster than c . We might try to remedy this by noting

that the Laplacian operator $\vec{\nabla}^2$ in (7.1) is equivalent to minus the d'Alembertian operator $\square^2$ in the limit $c \rightarrow \infty$, and thus postulate the modified field equation

$$\square^2 \Phi = -4\pi G\rho.$$

However, this equation does *not* yield a consistent relativistic theory. It is still not Lorentz covariant, since the matter density ρ does not transform as a Lorentz scalar. We shall discuss the transformation properties of the matter density later.

In addition to the incompatibility of Newtonian gravity with special relativity, there is a second fundamental difference between the electromagnetic and gravitational forces. The equation of motion of a particle of *inertial* mass m_1 in a gravitational field is given by

$$\frac{d^2 \vec{x}}{dt^2} = -\frac{m_G}{m_1} \vec{\nabla} \Phi. \quad (7.2)$$

It is a well-established experimental fact, however, that the ratio m_G/m_1 appearing in the equation of motion is *the same for all particles*. By an appropriate choice of units one may thus arrange for this ratio to equal unity. In contrast, the ratio q/m_1 occurring in the equation of motion of a charged particle in an electromagnetic field is *not* the same for all particles. From (7.2), we thus see that the trajectory through space of a particle in a gravitational field is *independent of the nature of the particle*.

This equivalence of the gravitational and inertial masses (which allows us to refer simply to 'the mass'), is a *truly remarkable* coincidence in the Newtonian theory. In this theory, there is no a-priori reason why the quantity that determines the magnitude of the gravitational force on the particle should equal the quantity that determines the particle's 'resistance' to an applied force in general. It appears as an isolated experimental result, which has since been verified to an accuracy of at least one part in 10^{11} (by Dicke and co-workers).

7.2 The equivalence principle

The equality of the gravitational and inertial masses of a particle led Einstein to his classic 'elevator' thought experiment. Consider an observer in a freely falling elevator (i.e. after the lift cable has been cut). Objects released from rest relative to the elevator cabin remain floating 'weightless' in the cabin. A projectile shot from one side of the elevator to the other appears to move in a straight line at constant velocity, rather than in the usual curved trajectory. All this follows from the fact that the acceleration of any particle relative to

the elevator is zero: the particle and the elevator cabin have the *same* acceleration relative to the Earth as a result of the equivalence of gravitational and inertial mass.

All these observations would hold *exactly* if the gravitational field of the Earth were truly *uniform*. Of course, the gravitational field of the Earth is not uniform but acts radially inwards towards its centre of mass, with a strength proportional to $1/r^2$. Thus, if the elevator were left to free-fall for a long time or if it were very large (i.e. a significant fraction of the Earth's radius), two particles released from rest near the walls of the elevator would gradually drift inwards, since they would both be falling along radial lines towards the centre of the Earth (see Figure 7.1). Furthermore, as a result of the varying strength of the gravitational field, particles released from rest near the floor of the elevator would gradually drift downwards whereas those near the ceiling would drift upwards. What the observer in the elevator would be experiencing would be the *tidal forces* resulting from the residual inhomogeneity in the strength and direction of the gravitational field once the main acceleration has been subtracted. It should always be remembered that these tidal forces can never be completely abolished in an elevator (laboratory) of finite, i.e. non-zero, size.

Nevertheless, provided that we consider the elevator cabin over a *short time period* and that it is *spatially small*, then a freely falling elevator (which may have (x, y, z) coordinates marked on its walls and an elevator clock measuring time t) resembles a Cartesian *inertial frame of reference*, and therefore *the laws of special relativity hold inside the elevator*.¹ These observations lead to

*The equivalence principle: In a freely falling (non-rotating) laboratory occupying a small region of spacetime, the laws of physics are those of special relativity.*²

7.3 Gravity as spacetime curvature

These observations led Einstein to make a profound proposal that simultaneously provides for a relativistic description of gravity *and* incorporates in a natural way the equivalence principle (and consequently the equivalence of gravitational and inertial mass). Einstein's proposal was that *gravity should no longer be regarded as a force in the conventional sense but rather as a manifestation of the curvature of the spacetime, this curvature being induced by the presence of matter*. This is the central idea underpinning the theory of general relativity.

¹ The elevator cabin must not only occupy a small region of spacetime but also be non-rotating with respect to distant matter in the universe. This statement is related to *Mach's principle*.

² This is in fact a statement of the *strong* equivalence principle, since it refers to all the Laws of physics. The more modest *weak* equivalence principle refers only to the trajectories of freely falling particles.

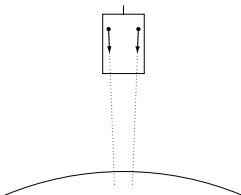


Figure 7.1 An elevator in free-fall towards the Earth.

If gravity is regarded a manifestation of the curvature of spacetime itself, and *not* as the action of some 4-force $\mathbf{f}$ defined on the manifold then the equation of motion of a particle moving only under the influence of gravity must be that of a ‘free’ particle in the curved spacetime, i.e.

$$\frac{d\mathbf{p}}{d\tau} = \mathbf{0},$$

where $\mathbf{p}$ is the particle’s 4-momentum and τ is the proper time measured along the particle’s worldline. Thus, *the worldline of a particle freely falling under gravity is a geodesic in the curved spacetime.*

The equivalence principle restricts the possible geometry of the curved spacetime to *pseudo-Riemannian*, as follows. The mathematical meaning of the equivalence principle is that it requires that *at any event P* in the spacetime manifold we must be able to define a coordinate system X^μ such that, *in the local neighbourhood of P* , the line element of spacetime takes the form

$$ds^2 \approx \eta_{\mu\nu} dX^\mu dX^\nu,$$

where exact equality holds at the event P . From the geodesic equation (as shown in Chapter 5), in such a coordinate system the path of a ‘free’ particle, i.e. one moving only under the influence of gravity, in the vicinity of the event P is given by

$$\frac{d^2 X^i}{dT^2} \approx 0,$$

where $i = 1, 2, 3$ and we have denoted X^0 by cT (once again the equality in the above equations holds exactly at P). Thus, in the vicinity of P the coordinates X^μ define a *local Cartesian inertial frame* (like our small elevator considered over a short time interval), in which the laws of special relativity hold locally. In order

that we can construct such a system, spacetime must be a pseudo-Riemannian manifold (which is curved and four-dimensional). For such a manifold, in some arbitrary coordinate system x^μ the line element takes the general form

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu.$$

7.4 Local inertial coordinates

The curvature of spacetime means that it is *not* possible to find coordinates in which the metric $g_{\mu\nu} = \eta_{\mu\nu}$ at all points in the manifold. Thus, it is not possible to define *global* Cartesian inertial frames as we could in the pseudo-Euclidean Minkowski spacetime. Instead, we are forced to use arbitrary coordinate systems x^μ to label events in spacetime, and these coordinates often do not have simple physical meanings. It is often the case that x^0 is a timelike coordinate and the $x^i (i = 1, 2, 3)$ are spacelike (i.e. the tangent vector to the x^0 coordinate curve is timelike at all points, and similarly the tangent vectors to the x^i coordinate curves are always spacelike). This allocation of coordinates is not necessary, however, and it is sometimes useful to define null coordinates. In any case, the arbitrary coordinates x^μ need not have any direct physical interpretation.

Nevertheless, as demanded by the equivalence principle, problems of physical meaning can always be overcome by transforming, at any event P in the curved spacetime, to a *local inertial coordinate system* X^μ , which, in a limited region of spacetime about P , corresponds to a freely falling, non-rotating, Cartesian frame over a short time interval. Mathematically, this corresponds to constructing about the event P a coordinate system X^μ such that

$$g_{\mu\nu}(P) = \eta_{\mu\nu} \quad \text{and} \quad (\partial_\sigma g_{\mu\nu})_P = 0. \quad (7.3)$$

This also means that $\Gamma^\mu_{\nu\sigma}(P) = 0$ and that the coordinate basis vectors at the event P form an orthonormal set, i.e.

$$\mathbf{e}_\mu(P) \cdot \mathbf{e}_\nu(P) = \eta_{\mu\nu}. \quad (7.4)$$

There are in fact an infinite number of local inertial coordinate systems at P , all of which are related to one another by Lorentz transformations. In other words, if a coordinate system X^μ satisfies the conditions (7.3), and hence the condition (7.4), then so too will the coordinate system

$$X'^\mu = \Lambda^\mu_{\nu} X^\nu,$$

where Λ^μ_{ν} defines a Lorentz transformation. Thus, local Cartesian freely falling (non-rotating) frames at an event P are related to one another by boosts, spatial

rotations or combinations of the two. For any one of these coordinate systems, the timelike basis vector $\mathbf{e}_0(P)$ is simply the normalised 4-velocity vector $\hat{\mathbf{u}}(P)$ of the origin of that frame at the event P , and the three mutually orthogonal spacelike vectors $\mathbf{e}_i(P)$ ($i = 1, 2, 3$) define the orientation of the spatial axes of the frame.

For points near to P , the metric in a local inertial coordinate system X^μ (whose origin is at P) is given by

$$g_{\mu\nu} = \eta_{\mu\nu} + \frac{1}{2}(\partial_\sigma \partial_\rho g_{\mu\nu})_P X^\sigma X^\rho + \dots$$

The sizes of the second derivatives $(\partial_\sigma \partial_\rho g_{\mu\nu})_P$ thus determine the region over which the approximation $g_{\mu\nu} \approx \eta_{\mu\nu}$ remains valid. We shall see the significance of these second derivatives shortly.

7.5 Observers in a curved spacetime

We discussed the subject of observers in Minkowski spacetime in Chapter 5, but let us now consider the subject in its full generality, in a curved spacetime. An observer will trace out some general (timelike) worldline $x^\mu(\tau)$ through spacetime, as expressed in some arbitrary coordinate system, where τ is the observer's proper time. An idealisation of his local laboratory is a frame of four orthonormal vectors $\hat{\mathbf{e}}_\alpha(\tau)$ (or tetrad) satisfying

$$\hat{\mathbf{e}}_\alpha(\tau) \cdot \hat{\mathbf{e}}_\beta(\tau) = \eta_{\alpha\beta},$$

which are carried with him along his worldline (these vectors may, in general, be totally unrelated to the basis vectors $\mathbf{e}_\mu$ of the coordinate system that we are using to label points in spacetime, although we can always express one set of vectors in terms of the other). In particular, at any point along his worldline the timelike vector $\hat{\mathbf{e}}_0(\tau)$ coincides with the normalised 4-velocity $\hat{\mathbf{u}}(\tau) = \mathbf{u}(\tau)/c$ of the observer. Similarly, the evolution of the spacelike vectors $\hat{\mathbf{e}}_i(\tau)$ along the worldline reflect the different ways in which his local laboratory may be spinning or tumbling. *Quantities measured in this laboratory correspond to projections of the relevant physical 4-vectors and 4-tensors onto this orthonormal frame.*

As shown in Chapter 5, if the observer has a 4-acceleration $\mathbf{a}(\tau) = d\mathbf{u}/d\tau$ but is not rotating, the tetrad basis vectors are Fermi–Walker-transported along the observer's worldline:

$$\frac{d\hat{\mathbf{e}}_\mu}{d\tau} = \frac{1}{c^2} [(\mathbf{u} \cdot \hat{\mathbf{e}}_\mu) \mathbf{a} - (\mathbf{a} \cdot \hat{\mathbf{e}}_\mu) \mathbf{u}]. \quad (7.5)$$

This expression holds equally well in a curved spacetime. An important special case is that of a non-rotating, *freely falling* observer, i.e. one who is moving only

under the influence of gravity. The vectors $\hat{e}_\alpha(\tau)$ then define what is called a *freely falling frame* (FFF). Free from any external forces, the observer's worldline traces out a geodesic in the curved spacetime. Thus the timelike vector $\hat{e}_0$ changes with proper time along the worldline according to

$$\frac{d\hat{e}_0}{d\tau} = \mathbf{0}.$$

In other words, $\hat{e}_0$ is parallel-transported along the worldline, and the observer's 4-acceleration $\mathbf{a}$ is zero. In this case we see from (7.5) that Fermi–Walker transport reduces to parallel transport. Thus the spacelike frame vectors $\hat{e}_i$ ($i = 1, 2, 3$) are also parallel-transported along the geodesic, so that

$$\frac{d\hat{e}_i}{d\tau} = \mathbf{0}.$$

Hence, in an arbitrary coordinate system x^μ , the components $(\hat{e}_\alpha)^\mu(\tau) = \hat{e}_\alpha(\tau) \cdot \mathbf{e}^\mu$ of any frame vector evolve as follows:

$$\frac{D(\hat{e}_\alpha)^\mu}{D\tau} = \frac{d(\hat{e}_\alpha)^\mu}{d\tau} + \Gamma^\mu_{\nu\sigma} (\hat{e}_\alpha)^\nu u^\sigma = 0.$$

This equation is extremely useful for determining what a freely falling observer would measure at a given event in spacetime. It is also clear that the frame vectors $\hat{e}_\alpha$ at any event P along the observer's worldline are the basis vectors of a local Cartesian inertial coordinate system at P .

7.6 Weak gravitational fields and the Newtonian limit

It is clear that, by construction, our description of gravity in terms of spacetime curvature reduces to special relativity in local inertial frames. It is important to check, however, that such a description also reduces to Newtonian gravity in the appropriate limits.

In the absence of gravity, spacetime has a Minkowski geometry. Therefore a weak gravitational field corresponds to a region of spacetime that is only 'slightly' curved. In other words, in such a region there exist coordinates x^μ in which the metric takes the form

$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \quad \text{where } |h_{\mu\nu}| \ll 1.$

(7.6)

Note that it is important to say 'there exist coordinates' since (7.6) does not hold for all coordinates; as we saw in Chapter 5, one can find coordinates even in Minkowski space in which $g_{\mu\nu}$ is not close to the simple form $\eta_{\mu\nu}$. Let us assume that in the coordinate system (7.6) the metric is *stationary*, which means that all

the derivatives $\partial_0 g_{\mu\nu}$ are zero. An example of such a coordinate system might be a fixed Cartesian frame at some point on the surface of the (non-rotating) Earth.

The worldline of a particle freely falling under gravity is given in general by the geodesic equation

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{\nu\sigma} \frac{dx^\nu}{d\tau} \frac{dx^\sigma}{d\tau} = 0.$$

We shall assume, however, that the particle is slow-moving, so that the components of its 3-velocity satisfy $dx^i/dt \ll c$ ($i = 1, 2, 3$), where t is defined by $x^0 \equiv ct$. This is equivalent to demanding that, for $i = 1, 2, 3$,

$$\frac{dx^i}{d\tau} \ll \frac{dx^0}{d\tau}.$$

Thus we can ignore the 3-velocity terms in the geodesic equation to obtain

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{00} c^2 \left(\frac{dt}{d\tau} \right)^2 = 0. \quad (7.7)$$

Now, recalling the expression (3.21) giving the connection in terms of the metric and using the form (7.6) for $g_{\mu\nu}$, we find that the connection coefficients Γ^μ_{00} are given by

$$\Gamma^\mu_{00} = \frac{1}{2} g^{\kappa\mu} (\partial_0 g_{0\kappa} + \partial_0 g_{0\kappa} - \partial_\kappa g_{00}) = -\frac{1}{2} g^{\kappa\mu} \partial_\kappa g_{00} = -\frac{1}{2} \eta^{\kappa\mu} \partial_\kappa h_{00},$$

where the last equality is valid to first order in $h_{\mu\nu}$. Since we have assumed that the metric is stationary, we have

$$\Gamma^0_{00} = 0 \quad \text{and} \quad \Gamma^i_{00} = \frac{1}{2} \delta^{ij} \partial_j h_{00},$$

where the Latin index runs over $i = 1, 2, 3$. Inserting these coefficients into (7.7) gives

$$\frac{d^2 t}{d\tau^2} = 0 \quad \text{and} \quad \frac{d^2 \vec{x}}{d\tau^2} = -\frac{1}{2} c^2 \left(\frac{dt}{d\tau} \right)^2 \vec{\nabla} h_{00}.$$

The first equation implies that $dt/d\tau = \text{constant}$, and so we can combine the two equations to yield the following equation of motion for the particle:

$$\frac{d^2 \vec{x}}{dt^2} = -\frac{1}{2} c^2 \vec{\nabla} h_{00}.$$

If we compare this equation with the usual *Newtonian* equation of motion for a particle in a gravitational field (7.2), we see that the two are *identical* if we make the identification $h_{00} = 2\Phi/c^2$. Hence for a slowly moving particle our

description of gravity as spacetime curvature tends to the Newtonian theory if the metric is such that, in the limit of a weak gravitational field,

$$g_{00} = \left(1 + \frac{2\Phi}{c^2}\right). \quad (7.8)$$

How big is the correction to the Minkowski metric? Some values of Φ/c^2 for various systems are as follows:

$$\frac{\Phi}{c^2} = -\frac{GM}{c^2 r} = \begin{cases} -10^{-9} & \text{at the surface of the Earth} \\ -10^{-6} & \text{at the surface of the Sun} \\ -10^{-4} & \text{at the surface of a white dwarf star.} \end{cases}$$

Thus, we see that even at the surface of a dense object like a white dwarf, the size of Φ/c^2 is much smaller than unity and hence the weak-field limit will be an excellent approximation.

From (7.8), the observant reader will have noticed that the description of gravity in terms of spacetime curvature has another immediate consequence, namely that the time coordinate t does not, in general, measure proper time. If we consider a clock at rest at some point in our coordinate system (i.e. $dx^i/dt = 0$), the proper time interval $d\tau$ between two ‘clicks’ of the clock is given by

$$c^2 d\tau^2 = g_{\mu\nu} dx^\mu dx^\nu = g_{00} c^2 dt^2,$$

from which we find that

$$d\tau = \left(1 + \frac{2\Phi}{c^2}\right)^{1/2} dt.$$

This gives the interval of proper $d\tau$ corresponding to an interval dt of coordinate time for a stationary observer near a massive object, in a region where the gravitational potential is Φ . Since Φ is negative, this proper time interval is shorter than the corresponding interval for a stationary observer at a large distance from the object, where $\Phi \rightarrow 0$ and so $d\tau = dt$. Thus, as a bonus, our analysis has also yielded the formula for *time dilation* in a weak gravitational field.

7.7 Electromagnetism in a curved spacetime

Before going on to discuss the mathematics of curvature in detail, let us look back at our development of electromagnetism in Chapter 6. It is clear that our derivation of the electromagnetic field equations in arbitrary coordinates did *not*

depend on the intrinsic geometry of the manifold on which the electromagnetic field tensor F and the 4-current j are defined. In other words, one can arrive at these equations without assuming the spacetime to have a Minkowski geometry. Thus, in the presence of gravitating matter, spacetime becomes curved but the field equations of electromagnetism in an arbitrary coordinate system are still given by

$$\begin{aligned} \nabla_\mu F^{\mu\nu} &= \mu_0 j^\nu, \\ \nabla_\sigma F_{\mu\nu} + \nabla_\nu F_{\sigma\mu} + \nabla_\mu F_{\nu\sigma} &= 0. \end{aligned} \quad (7.9)$$

The effects of gravitation are automatically included in these field equations through the covariant derivatives, which depend on the metric $g_{\mu\nu}$ describing the spacetime geometry. Moreover, if we construct a local Cartesian coordinate system about some point P in the manifold then (as discussed above) these coordinates correspond to a local inertial frame in the neighbourhood of P . In these coordinates, the equations of electromagnetism then take their familiar special relativistic forms.

An electromagnetic field tensor F defined on a curved spacetime gives rise (as in Minkowski space) to a 4-force $f = qF \cdot u$, which acts on a particle of charge q with 4-velocity u . Thus the equation of motion of a charged particle moving under the influence of an electromagnetic field in a curved spacetime has the same form as that in Minkowski spacetime, i.e.

$$m_0 \frac{du}{d\tau} = qF \cdot u,$$

where m_0 is the rest mass of the particle. In this case, however, because of the curvature of spacetime the particle is moving under the influence of both electromagnetic forces *and* gravity. In some arbitrary coordinate system, the particle's worldline is again given by

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{\nu\sigma} \frac{dx^\nu}{d\tau} \frac{dx^\sigma}{d\tau} = \frac{q}{m_0} F^\mu_{\nu} \frac{dx^\nu}{d\tau}.$$

Obviously, in the absence of an electromagnetic field (or for an uncharged particle), the right-hand side is zero and we recover the equation of a geodesic.

We must remember, however, that the energy and momentum of the electromagnetic field will *itself* induce a curvature of spacetime, so the metric in this case is determined not only by the matter distribution but also by the radiation.

7.8 Intrinsic curvature of a manifold

Since the notion of curvature is central to general relativity, we must now investigate how to quantify the intrinsic curvature of a manifold at any given point P .³ A manifold (or region of a manifold) is *flat* if there exist coordinates X^μ such that, throughout the region, the line element can be written

$$ds^2 = \epsilon_1 (dX^1)^2 + \epsilon_2 (dX^2)^2 + \cdots + \epsilon_N (dX^N)^2, \quad (7.10)$$

where $\epsilon_a = \pm 1$ (in other words ‘flat’ is a shorthand for pseudo-Euclidean). If, however, points in the manifold are labelled with some arbitrary coordinate system x^a then in general the line element ds^2 will not be of the above form. Thus, if for some manifold the line element is given by

$$ds^2 = g_{ab}(x) dx^a dx^b,$$

how can we tell whether the intrinsic geometry of the manifold in some region is flat or curved in some way?

Consider, for example, the following line element for a three-dimensional space:

$$ds^2 = dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2.$$

Of course, we recognise this as the line element of ordinary three-dimensional Euclidean space written in spherical polar coordinates. In other words, the transformation

$$x = r \sin \theta \cos \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta$$

will turn the above line element into the form

$$ds^2 = dx^2 + dy^2 + dz^2. \quad (7.11)$$

But what about other line elements? For example, recall from Chapter 2 the three-dimensional space described by the line element (2.21):

$$ds^2 = \frac{a^2}{a^2 - r^2} dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2.$$

How can we tell whether this metric, or a more complicated metric, corresponds to flat space but merely looks complicated because of a weird choice of coordinates? It would be immensely tedious to try to discover whether there exists a coordinate transformation that reduces a metric to the form (7.11). We therefore need some means of telling whether a manifold is flat directly from the metric g_{ab} , *independently of the coordinate system being used*.

³ Since the material presented here is applicable to *any* N -dimensional pseudo-Riemannian manifold, we will use indices a, b etc. that have a range 1 to N , rather than μ, ν etc., with a range 0 to 3. Of course, the final application to general relativity will govern the scope of our results.

The *physical* significance of this to general relativity is as follows. If, throughout some region of a four-dimensional spacetime, we can reduce the line element

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu$$

to Minkowski form then there can be no gravitational field in this region. The equivalence of a general line element to that of Minkowski spacetime therefore guarantees that the gravitational field will vanish. The solution to our mathematical problem of finding a coordinate-independent way of defining the curvature of spacetime will lead us to the field equations of gravity.

7.9 The curvature tensor

We can find a solution to the problem of measuring the curvature of a manifold at any point by considering changing the order of covariant differentiation. Covariant differentiation is clearly a generalisation of partial differentiation. There is one important respect in which it differs, however: it matters in which order covariant differentiation is performed, and changing the order (in general) changes the result.

Since for a scalar field the covariant derivative is simply the partial derivative, the order of differentiation does not matter. However, let us consider some arbitrary vector field defined on a manifold, with covariant components v_a . The covariant derivative of the v_a is given by

$$\nabla_b v_a = \partial_b v_a - \Gamma^d_{ab} v_d.$$

A second covariant differentiation then yields

$$\begin{aligned} \nabla_c \nabla_b v_a &= \partial_c (\nabla_b v_a) - \Gamma^e_{ac} \nabla_b v_e - \Gamma^e_{bc} \nabla_e v_a \\ &= \partial_c \partial_b v_a - (\partial_c \Gamma^d_{ab}) v_d - \Gamma^d_{ab} \partial_c v_d \\ &\quad - \Gamma^e_{ac} (\partial_b v_e - \Gamma^d_{eb} v_d) - \Gamma^e_{bc} (\partial_e v_a - \Gamma^d_{ae} v_d), \end{aligned}$$

which follows since $\nabla_b v_a$ is itself a rank-2 tensor. Swapping the indices b and c to obtain a corresponding expression for $\nabla_b \nabla_c v_a$ and then subtracting gives

$$\nabla_c \nabla_b v_a - \nabla_b \nabla_c v_a = R^d_{abc} v_d, \quad (7.12)$$

where

$$R^d_{abc} \equiv \partial_b \Gamma^d_{ac} - \partial_c \Gamma^d_{ab} + \Gamma^e_{ac} \Gamma^d_{eb} - \Gamma^e_{ab} \Gamma^d_{ec}. \quad (7.13)$$

To determine directly whether the N^4 quantities R^d_{abc} transform as the components of a tensor under a coordinate transformation would be an arduous algebraic task. Fortunately the quotient theorem (Section 4.11) provides a much shorter route. The left-hand side of (7.12) is a tensor, for arbitrary vectors v_a , so the

contraction of $R^d{}_{abc}$ with v_d is also a tensor. Since $R^d{}_{abc}$ does not depend on v_a , we conclude from the quotient theorem that the $R^d{}_{abc}$ are indeed the components of some rank-4 tensor $\mathbf{R}$. This tensor is called the *curvature tensor* (or *Riemann tensor*), and equation (7.13) shows that it is defined in terms of the metric tensor g_{ab} and its first and second derivatives.

We must now establish how the tensor (7.13) is related to the curvature of the manifold. In a *flat* region of a manifold, we may choose coordinates such that the line element takes the form (7.10) throughout the region. In these coordinates $\Gamma^a{}_{bc}$ and its derivatives are zero, and hence

$$R^d{}_{abc} = 0$$

at every point in the region. This is a tensor relation, however, and so it must hold in any coordinate system. Conversely, if $R^d{}_{abc} = 0$ at every point in some region of a manifold, then it may be shown that it is possible to introduce a coordinate system in which the line element takes the form (7.10), and hence this region is *flat*.⁴ Thus the vanishing of the curvature tensor is a necessary and sufficient condition for a region of a manifold to be flat.

7.10 Properties of the curvature tensor

The curvature tensor (7.13) possesses a number of symmetries and satisfies certain identities, which we now discuss. The symmetries of the curvature tensor are most easily derived in terms of its covariant components

$$R_{abcd} = g_{ae} R^e{}_{bcd}.$$

For completeness, we note that in an arbitrary coordinate system an explicit form for these components is found, after considerable algebra, to be

$$R_{abcd} = \frac{1}{2}(\partial_d \partial_a g_{bc} - \partial_d \partial_b g_{ac} + \partial_c \partial_b g_{ad} - \partial_c \partial_a g_{bd}) - g^{ef}(\Gamma_{eac} \Gamma_{fbd} - \Gamma_{ead} \Gamma_{fbc}).$$

One could use this expression straightforwardly to derive the symmetry properties of the curvature tensor, but we take the opportunity here to illustrate a general mathematical device that is often useful in reducing the algebraic burden of tensor manipulations.

Let us choose some arbitrary point P in the manifold and construct a geodesic coordinate system about this point (see Section 3.11), in which the connection vanishes, $\Gamma^a{}_{bc}(P) = 0$, although in general its derivatives will not. In this

⁴ For a proof of this result, see (for example) P. A. M. Dirac, *General Theory of Relativity*, Princeton Landmarks in Physics Series, Princeton University Press, 1996.

coordinate system, one may easily show directly from (7.13) that the covariant components of the curvature tensor at P are given by

$$(R_{abcd})_P = \frac{1}{2}(\partial_d \partial_a g_{bc} - \partial_d \partial_b g_{ac} + \partial_c \partial_b g_{ad} - \partial_c \partial_a g_{bd})_P.$$

From this expression one may immediately establish the following symmetry properties at P :

$$R_{abcd} = -R_{bacd}, \quad (7.14)$$

$$R_{abcd} = -R_{abdc}, \quad (7.15)$$

$$R_{abcd} = R_{cdab}. \quad (7.16)$$

The first two properties show that the curvature tensor is antisymmetric with respect to swapping the order of either the first two indices or the second two indices. The third property shows that it is symmetric with respect to swapping the first *pair* of indices with the second *pair* of indices. Moreover, we may also easily deduce the *cyclic identity*

$$R_{abcd} + R_{acdb} + R_{adbc} = 0, \quad (7.17)$$

which on using (7.15) may be written more succinctly as $R_{a[bcd]} = 0$. Although the results (7.14–7.17) have been derived in a special coordinate system, each condition is a tensor relation and so if it is valid in one coordinate system then it is valid in all. Moreover, since the point P is arbitrary, the results hold everywhere.

Although first appearances might suggest that the curvature tensor has N^4 components, the conditions (7.14–7.17) reduce the number of independent components to $N^2(N^2 - 1)/12$. Recall from Section 2.11 that this is also the number of degrees of freedom among the second derivatives $\partial_d \partial_c g_{ab}$. This is not surprising since, at any point P in a manifold, we can perform a transformation to local Cartesian coordinates in which $g_{ab}(P) = \eta_{\mu\nu}$ and $(\partial_c g_{ab})_P = 0$. Thus, a general metric at any point P is characterised by the $N^2(N^2 - 1)/12$ second derivatives that cannot be made to vanish there.

For manifolds of different dimensions we have the following results:

No. of dimensions	2	3	4
No. of independent components of R_{abcd}	1	6	20

You can see from this table that in four dimensions the number of independent components is reduced from a possible 256 to 20. You will also see that in one dimension the curvature tensor is always equal to zero: $R_{1111} = 0$. How can this

be? Can a line not be curved? Think about this – the curvature measures the ‘inner’ properties of the space. When we say that a line is curved we refer to a particular embedding in a higher-dimensional space, but this does not tell us about the inner properties of the space. In one dimension, it is evident that we can *always* find a coordinate transformation that will reduce an arbitrary metric to the form (7.10). As a two-dimensional example, in Appendix 7A we calculate the single independent component of the curvature tensor for the surface of a sphere. The *Gaussian curvature* K of a two-dimensional surface is given by

$$K = \frac{R_{1212}}{g},$$

where $g = \det[g_{ab}]$ is the determinant of the metric tensor.

The curvature tensor also satisfies a differential identity, which may be derived as follows. Let us once again adopt a geodesic coordinate system about some arbitrary point P . In this coordinate system, differentiating and then evaluating the result at P gives

$$(\nabla_e R_{abcd})_P = (\partial_e R_{abcd})_P = (\partial_e \partial_c \Gamma_{abd} - \partial_e \partial_d \Gamma_{abc})_P.$$

Cyclically permuting c , d and e to obtain two further analogous relations and adding, one finds that at P

$$\boxed{\nabla_e R_{abcd} + \nabla_c R_{abde} + \nabla_d R_{abec} = 0.} \quad (7.18)$$

This is, however, a tensor relation and thus holds in all coordinate systems; moreover, since P is arbitrary the relationship holds everywhere. This result is known as the *Bianchi identity* and, using the antisymmetry relation (7.14), it may be written more succinctly as

$$\nabla_{[e} R_{ab]cd} = 0.$$

7.11 The Ricci tensor and curvature scalar

It follows from the symmetry properties (7.14–7.16) of the curvature tensor that it possesses only two independent contractions. We may find these by contracting either on the first two indices or on the first and last indices respectively. From (7.14), raising the index a and then contracting on the first two indices gives

$$R^a_{acd} = 0.$$

Contracting on the first and last indices, however, gives in general a non-zero result and this leads to a new tensor, the *Ricci tensor*. It is traditional to use the

same kernel letter for the Ricci tensor as for the curvature tensor, so we denote its components by

$$R_{ab} \equiv R^c{}_{abc}.$$

By raising the index a in the cyclic identity (7.17) and contracting with d , one may easily show that the Ricci tensor is symmetric. Thus we have $R^a{}_b = R_a{}^b$ and we can denote both by $R^b{}_a$.

A further contraction gives the *curvature scalar* (or *Ricci scalar*)

$$R \equiv g^{ab} R_{ab} = R^a{}_a,$$

where again the same kernel letter is used. This is a scalar quantity defined at each point of the manifold.

The covariant derivatives of the Ricci tensor and the curvature scalar obey a particularly important relation, which will be central to our development of the field equations of general relativity. Raising a in the Bianchi identity (7.18) and contracting with d gives

$$\nabla_e R_{bc} + \nabla_c R^a{}_{bae} + \nabla_a R^a{}_{bec} = 0,$$

which, on using the antisymmetry property (7.16) in the second term, gives

$$\nabla_e R_{bc} - \nabla_c R_{be} + \nabla_a R^a{}_{bec} = 0.$$

If we now raise b and contract with e , we find

$$\nabla_b R^b{}_c - \nabla_c R + \nabla_a R^{ab}{}_{bc} = 0. \quad (7.19)$$

Using the antisymmetry properties (7.14, 7.15) we may write the third term as

$$\nabla_a R^{ab}{}_{bc} = \nabla_a R^{ba}{}_{cb} = \nabla_a R^a{}_c = \nabla_b R^b{}_c,$$

so the first and last terms in (7.19) are identical and we obtain

$$2\nabla_b R^b{}_c - \nabla_c R = \nabla_b (2R^b{}_c - \delta^b{}_c R) = 0.$$

Finally, raising the index c , we obtain the important result

$$\nabla_b (R^{bc} - \frac{1}{2}g^{bc}R) = 0.$$

The term in parentheses is called the *Einstein tensor*

$$G^{ab} \equiv R^{ab} - \frac{1}{2}g^{ab}R.$$

It is clearly symmetric and thus possesses only one independent divergence $\nabla_a G^{ab}$, which vanishes (by construction). As we will see, it is this tensor that describes the curvature of spacetime in the field equations of general relativity.

7.12 Curvature and parallel transport

In Chapter 3, we remarked that parallel transport in a curved manifold was path dependent. We now have a more formal description of curvature. If a region of manifold is flat then the curvature tensor vanishes throughout the region; otherwise, it is curved. Thus there must be some link between the curvature tensor and parallel transport.

Let us consider the parallel transport of a vector v around a closed curve $\mathcal{C}$ in a manifold. We can define an arbitrary surface $\mathcal{A}$ bounding the curve $\mathcal{C}$ and break this surface up into a lot of small areas each bounded by closed curves $\mathcal{C}_N$, as indicated in Figure 7.2. The change in the components v^a on being parallel-transported around the closed curve $\mathcal{C}$ is then

$$\Delta v^a = \sum_N (\Delta v^a)_N,$$

where $(\Delta v^a)_N$ is the change in v^a around the small closed curve $\mathcal{C}_N$. This follows because the changes in Δv^a around any of the interior closed curves cancel, leaving just the contributions around the outer edges that bound the curve $\mathcal{C}$.

Let us now calculate $(\Delta v^a)_N$ around the small closed curve $\mathcal{C}_N$ defined by the parametric equations $x^a(u)$. The equation for parallel transport is given by (3.41):

$$\frac{dv^a}{du} = -\Gamma^a_{bc} v^b \frac{dx^c}{du}.$$

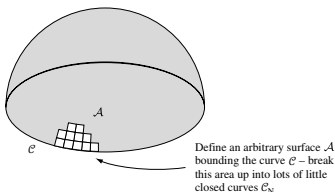


Figure 7.2 An arbitrary surface $\mathcal{A}$ bounding a closed curve $\mathcal{C}$.

Thus, if v^a is parallel-transported along the small closed curve $\mathcal{C}_N$ from some initial point P then at some other point along this curve we have

$$v^a(u) = v_P^a - \int_{u_P}^u \Gamma_{bc}^a v_P^b \frac{dx^c}{du} du. \quad (7.20)$$

However, since the closed curve is small we can expand the factors in the integrand about P to first order in $x^a - x_P^a$:

$$\begin{aligned} \Gamma_{bc}^a(u) &= (\Gamma_{bc}^a)_P + (\partial_d \Gamma_{bc}^a)_P [x^d(u) - x_P^d] + \cdots, \\ v^a(u) &= v_P^a - (\Gamma_{bc}^a)_P v_P^b [x^c(u) - x_P^c] + \cdots. \end{aligned}$$

Substituting these expressions into (7.20) and retaining terms only up to first order in $x^a - x_P^a$, we obtain

$$\begin{aligned} v^a(u) &= v_P^a - (\Gamma_{bc}^a)_P v_P^b \int_{u_P}^u \frac{dx^c}{du} du \\ &\quad - (\partial_d \Gamma_{bc}^a - \Gamma_{ec}^a \Gamma_{bd}^e)_P v_P^b \int_{u_P}^u (x^d - x_P^d) \frac{dx^c}{du} du. \end{aligned}$$

If we integrate the coordinate differentials around a closed loop we have $\oint dx^c = 0$, and so we find that

$$\Delta v^a = -(\partial_d \Gamma_{bc}^a - \Gamma_{ec}^a \Gamma_{bd}^e)_P v_P^b \oint_{u_P}^u x^d dx^c.$$

We may obtain an analogous result by interchanging the dummy indices c and d . Now using the result

$$\oint d(x^c x^d) = \oint (x^c dx^d + x^d dx^c) = 0,$$

we find that

$$\Delta v^a = -\frac{1}{2} (\partial_c \Gamma_{bd}^a - \partial_d \Gamma_{bc}^a + \Gamma_{ec}^a \Gamma_{bd}^e - \Gamma_{ed}^a \Gamma_{bc}^e)_P v_P^b \oint x^c dx^d.$$

On using the expression (7.13), we finally obtain

$$\Delta v^a = -\frac{1}{2} (R_{bcd}^a)_P v_P^b \oint x^c dx^d.$$

(7.21)

Equation (7.21) establishes the link between the curvature tensor at a point P and parallel transport around a small loop close to P . It tells us that the components v^a will remain unchanged after parallel transportation around a small closed loop near P if and only if the curvature tensor vanishes at P . So, returning

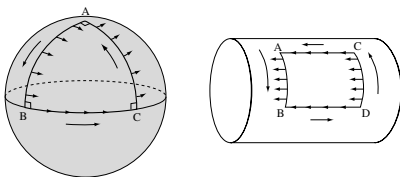


Figure 7.3 Parallel transport around a closed curve on the surface of a sphere and the surface of a cylinder.

to our construction of $(\Delta v^a)_N$, the vector components v^a will not change on parallel transportation around the entire closed curve $\mathcal{C}$ if the curvature tensor R^a_{bcd} vanishes over the entire area $\mathcal{A}$ bounding the curve.

As an example, consider the parallel transportation of a vector around the closed triangle ABC on the surface of a sphere (see Figure 7.3). As shown in Appendix 7A, the curvature tensor is nowhere zero, and it is evident that the vector changes direction after parallel transportation around the triangle. However, as also mentioned in Appendix 7A, the curvature tensor vanishes everywhere on the surface of a cylinder and hence the components of a vector will remain unchanged if the vector is parallel-transported around any closed curve (see Figure 7.3).

7.13 Curvature and geodesic deviation

Another important consequence of curvature is that two nearby geodesics that are initially parallel either converge or diverge, depending on the local curvature. This is embodied in the *equation of geodesic deviation*, which we now derive.

Consider two neighbouring geodesics, $\mathcal{C}$ given by $x^a(u)$ and $\tilde{\mathcal{C}}$ given by $\tilde{x}^a(u)$, where u is an affine parameter, and let $\xi^a(u)$ be the small ‘vector’ connecting points on the two geodesics with the same parameter value (see Figure 7.4), i.e.

$$\tilde{x}^a(u) = x^a(u) + \xi^a(u).$$

In particular, let us suppose that for some arbitrary value of u the vector $\xi^a(u)$ connects the point P on $\mathcal{C}$ to the point Q on $\tilde{\mathcal{C}}$.

Once again our derivation is simplified considerably by constructing local geodesic coordinates about the point P , in which the connection coefficients

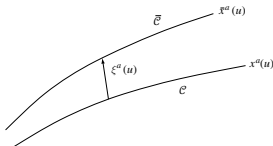


Figure 7.4 Two neighbouring geodesics.

vanish at P but their derivatives are in general non-zero there. In this coordinate system, since $\mathcal{C}$ and $\bar{\mathcal{C}}$ are geodesics we have

$$\left(\frac{d^2 x^a}{du^2} \right)_P = 0, \quad (7.22)$$

$$\left(\frac{d^2 \bar{x}^a}{du^2} + \Gamma^a_{bc} \frac{d\bar{x}^b}{du} \frac{d\bar{x}^c}{du} \right)_Q = 0, \quad (7.23)$$

at the points P and Q respectively. However, to first order in ξ^a ,

$$\Gamma^a_{bc}(Q) = \Gamma^a_{bc}(P) + (\partial_d \Gamma^a_{bc})_P \xi^d = (\partial_d \Gamma^a_{bc})_P \xi^d.$$

Thus, subtracting (7.22) from (7.23) gives, to first order, at P

$$\ddot{\xi}^a + (\partial_d \Gamma^a_{bc}) \dot{x}^b \dot{x}^c \xi^d = 0,$$

where the dots denote d/du . However, in our geodesic coordinates the second-order intrinsic derivative of ξ^a at P is given by

$$\frac{D^2 \xi^a}{Du^2} = \frac{d}{du} \left(\dot{\xi}^a + \Gamma^a_{bc} \xi^b \dot{x}^c \right) = \ddot{\xi}^a + (\partial_d \Gamma^a_{bc}) \xi^b \dot{x}^c \dot{x}^d,$$

where we have used the fact that $\Gamma^a_{bc}(P) = 0$; we note that nevertheless the derivatives of Γ^a_{bc} at P may not vanish. Thus, combining the last two equations and relabelling dummy indices, we find that at P

$$\frac{D^2 \xi^a}{Du^2} + (\partial_b \Gamma^a_{cd} - \partial_d \Gamma^a_{bc}) \xi^b \dot{x}^c \dot{x}^d = 0.$$

We may now identify the terms in parentheses on the left-hand side as components R^a_{cbd} of the Riemann tensor when expressed in local geodesic coordinates about P . Thus we may write the above result as

$$\frac{D^2 \xi^a}{Du^2} + R^a_{cbd} \xi^b \dot{x}^c \dot{x}^d = 0,$$

(7.24)

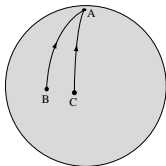


Figure 7.5 Converging geodesics on the surface of a sphere.

which is clearly a tensor relation and is hence valid in *any* coordinate system. Moreover, since P is an arbitrary point on $\mathcal{C}$, this relation is valid everywhere along the curve. The result (7.24) is the *equation of geodesic deviation*.

The geometric meaning of (7.24) is straightforward. In a flat region of a manifold, $R^a_{bcd} = 0$ and we may adopt Cartesian coordinates throughout. In this case, $D/Du = d/du$ and the equation of geodesic deviation reduces to $d^2\xi^a/du^2 = 0$, which implies that $\xi^a(u) = A^au + B^a$ where A^a and B^a are constants. So in a flat region the separation vector $\xi^a(u)$ connecting the two geodesics (which are simple straight lines in this case) in general increases *linearly* with u . In the special case where the two lines are initially parallel then they will remain so and hence never intersect. In a curved region of a manifold, however $R^a_{bcd} \neq 0$ and so neighbouring geodesics either converge or diverge. For example, the two neighbouring geodesics AB and AC on the surface of a sphere (see Figure 7.5) converge as we approach the point A at the pole because the surface is positively curved. Equation (7.24) allows us to compute the rates of convergence or divergence of neighbouring geodesics for Riemannian spaces of arbitrary complexity. All one needs to do is to compute the curvature tensor (7.13) at each point using the metric.

7.14 Tidal forces in a curved spacetime

Now that we have derived the equation of geodesic variation (7.24), we can give a more quantitative account of the *gravitational tidal forces* mentioned in our discussion of the equivalence principle in Section 7.2. Let us begin by working in Newtonian gravity and consider an initially spherical distribution of non-interacting particles freely falling towards the Earth (see Figure 7.6). Each particle moves on a straight line through the centre of the Earth, but those nearer the Earth fall faster because the gravitational attraction is stronger. Thus the sphere no longer remains a sphere but is distorted into an ellipsoid of the same

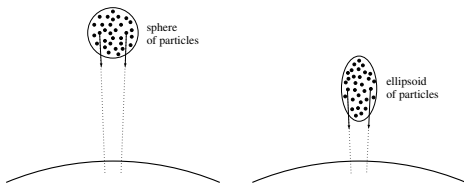


Figure 7.6 Tidal force on a collection of non-interacting particles.

volume: gravity has produced a tidal force in the sphere of particles that results in an elongation of the distribution in the direction of motion and a compression of the distribution in the transverse directions. Indeed, it is straightforward to show that, for two nearby particles with trajectories $x^i(t)$ and $\bar{x}^i(t)$ ($i = 1, 2, 3$) respectively in Cartesian coordinates, that the components of the separation vector $\zeta^i = x^i - \bar{x}^i$ evolve as

$$\frac{d^2 \zeta^i}{dt^2} = - \left(\frac{\partial^2 \Phi}{\partial x^i \partial x^j} \right) \zeta^j,$$

where Φ is the Newtonian gravitational potential (see Exercise 7.21).

A similar tidal effect occurs in general relativity and can be understood in terms of the curvature of the spacetime. In particular, we can gain some idea of the general-relativistic tidal forces by considering the equation of geodesic deviation (7.24). Consider any pair of our non-interacting particles. Each one is in free fall and so they must move along the timelike geodesics $x^\mu(\tau)$ and $\bar{x}^\mu(\tau)$ respectively, where τ is the proper time experienced by the first particle (say). If we define a small separation vector between the two particle worldlines by $\xi^\mu(\tau) = \bar{x}^\mu(\tau) - x^\mu(\tau)$, then (7.24) shows that it evolves according to the equation

$$\boxed{\frac{D^2 \xi^\mu}{D\tau^2} = S^\mu{}_\nu \xi^\nu}, \quad (7.25)$$

where we have defined the *tidal stress tensor*

$$\boxed{S^\mu{}_\nu \equiv R^\mu{}_{\sigma\rho\nu} u^\sigma u^\rho}, \quad (7.26)$$

in which $u^\sigma \equiv dx^\sigma/d\tau$ is the 4-velocity of the first particle. Note that in defining $S^\mu{}_\nu$ we have made use of the fact that the curvature tensor is antisymmetric in its last two indices. The result (7.25) is a fully covariant tensor equation and therefore holds in any coordinate system.

To understand the physical consequences of the geodesic deviation effect, it is helpful to consider how some observer will view the relative spatial acceleration of the two particles. Suppose that our observer is sitting on the first particle, the worldline $x^\mu(\tau)$ of which passes through some event P . In order to calculate the relative spatial acceleration measured by our observer, we may erect a set of orthonormal basis vectors $\hat{e}_\alpha$ at P that define the instantaneous rest frame (IRF) of the first particle (and the observer) at this event. The timelike basis vector is given simply by $\hat{e}_0 = \hat{u}$, where u is the 4-velocity at P of the first particle, and we may choose the spacelike basis vectors $\hat{e}_i$ in any way, provided that the full set satisfies

$$\hat{e}_\alpha \cdot \hat{e}_\beta = \eta_{\alpha\beta}.$$

In this way, the duals of these basis vectors, which are given by $\hat{e}^\alpha = \eta^{\alpha\beta} \hat{e}_\beta$, also form an orthonormal set. The general situation is illustrated schematically in Figure 7.7.

The components of the separation vector ξ with respect to our new frame are

$$\xi^{\hat{\alpha}} \equiv \hat{e}^\alpha \cdot \xi = (\hat{e}^\alpha)_\mu \xi^\mu;$$

these components give the temporal and spatial separations of the events P and Q on the two particle worldlines, as measured by our observer. Since the $\hat{e}_\alpha$ ($\alpha = 0, 1, 2, 3$) are the basis vectors of an inertial Cartesian coordinate system at P , the intrinsic derivative in this coordinate system is simply equal to the ordinary

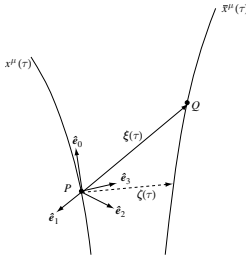


Figure 7.7 Schematic illustration of the basis vectors of the instantaneous rest frame at P . A general connecting vector ξ and the orthogonal connecting vector ζ are also shown.

derivative. Moreover, with respect to the IRF, the 4-velocity of the first particle is simply $[u^{\hat{\alpha}}] = (c, \vec{0})$. Thus from (7.25) we have

$$\frac{d^2 \xi^{\hat{\alpha}}}{d\tau^2} = c^2 R^{\hat{\alpha}}{}_{\hat{0}\hat{0}\hat{\gamma}} \xi^{\hat{\gamma}}, \quad (7.27)$$

where the components of the curvature tensor in the Cartesian inertial frame at P may be written as

$$R^{\hat{\alpha}}{}_{\hat{\beta}\hat{\gamma}\hat{\delta}} \equiv R^{\mu}{}_{\sigma\nu\rho} (\hat{e}^{\alpha})_{\mu} (\hat{e}_{\beta})^{\sigma} (\hat{e}_{\gamma})^{\nu} (\hat{e}_{\delta})^{\rho}. \quad (7.28)$$

Equation (7.27) in fact holds for any orthonormal freely falling frame $\hat{e}_{\alpha}$.

Clearly, the general separation vector ξ is inappropriate for our discussion of the evolution of the *spatial* separation seen by our observer at P , since typically ξ will have some temporal component in the observer's frame. Thus, we must work instead with the *orthogonal connecting vector* $\zeta(\tau)$ shown in Figure 7.7, which has a zero component in the $\hat{e}_0$ -direction, i.e. $\zeta^{\hat{0}} = 0$. Since (7.27) is valid for any small connecting vector it must also hold for the orthogonal connecting vector ζ , but we must remember that $\zeta^{\hat{0}}(\tau) = 0$ for all τ .

A useful alternative interpretation of (7.25) or (7.27) is that it gives the force per unit mass required to keep two particles moving along parallel curves; this force must be supplied by some mechanical means. For example, the worldline of the centre of mass of a rigid body in free fall is a timelike geodesic, but this is not true of the other parts of the object, which are constrained to move along curves parallel to the centre of mass rather than along neighbouring geodesics. The necessary forces must be supplied by internal stresses in the object. The physical magnitude of the stresses is most easily found by solving the eigenvalue problem

$$S^{\mu}{}_{\nu} v^{\nu} = \lambda v^{\mu}, \quad (7.29)$$

where $S^{\mu}{}_{\nu}$ is given by (7.26). One of the eigenvalues is always zero (for $v^{\mu} = u^{\mu}$), and the remaining three eigenvalues give the *principal stresses* in the object.

Appendix 7A: The surface of a sphere

The metric⁵ of the surface of a sphere in spherical polar coordinates is

$$ds^2 = a^2 d\theta^2 + a^2 \sin^2 \theta d\phi^2.$$

⁵ Note that this term is often applied, as here, to the line element itself.

To get used to handling problems involving curved spaces you should calculate the components of the affine connection, starting from this metric. The definition of the affine connection is

$$\Gamma_{bc}^a = \frac{1}{2}g^{ad}(\partial_b g_{dc} + \partial_c g_{bd} - \partial_d g_{bc}),$$

as given in (3.21), and in two dimensions there are six independent connection coefficients,

$$\Gamma_{11}^1, \quad \Gamma_{12}^1, \quad \Gamma_{22}^1, \quad \Gamma_{11}^2, \quad \Gamma_{12}^2, \quad \Gamma_{22}^2.$$

These coefficients are given by (exercise):

$$\begin{aligned}\Gamma_{11}^1 &= \frac{1}{2}g^{11}(\partial_1 g_{11} + \partial_1 g_{11} - \partial_1 g_{11}) = 0, \\ \Gamma_{12}^1 &= \frac{1}{2}g^{11}(\partial_2 g_{11} + \partial_2 g_{21} - \partial_1 g_{12}) = 0, \\ \Gamma_{22}^1 &= \frac{1}{2}g^{11}(\partial_2 g_{21} + \partial_2 g_{21} - \partial_1 g_{22}) = -\frac{1}{2}g^{11}\partial_1 g_{22}, \\ \Gamma_{11}^2 &= \frac{1}{2}g^{22}(\partial_2 g_{12} + \partial_2 g_{12} - \partial_2 g_{11}) = 0, \\ \Gamma_{12}^2 &= \frac{1}{2}g^{22}(\partial_1 g_{22} + \partial_2 g_{12} - \partial_2 g_{11}) = \frac{1}{2}g^{22}\partial_1 g_{22}, \\ \Gamma_{22}^2 &= \frac{1}{2}g^{22}(\partial_2 g_{22} + \partial_2 g_{22} - \partial_2 g_{22}) = 0.\end{aligned}$$

So, the only two non-zero coefficients are

$$\begin{aligned}\Gamma_{22}^1 &= -\frac{1}{2a^2}2a^2 \sin \theta \cos \theta = -\sin \theta \cos \theta, \\ \Gamma_{12}^2 &= \frac{1}{2a^2 \sin^2 \theta}2a^2 \sin \theta \cos \theta = \frac{\cos \theta}{\sin \theta}.\end{aligned}$$

The curvature tensor is

$$R_{abcd} = \frac{1}{2}(\partial_d \partial_a g_{bc} - \partial_d \partial_b g_{ac} + \partial_c \partial_b g_{ad} - \partial_c \partial_a g_{bd}) - g_{ef}(\Gamma_{ac}^e \Gamma_{bd}^f - \Gamma_{ad}^e \Gamma_{bc}^f)$$

and in two dimensions the symmetry properties of this tensor mean that there is only *one* independent component. We can take this to be R_{1212} , so fortunately we only have to calculate this single component:

$$\begin{aligned}R_{1212} &= \frac{1}{2}(\partial_2 \partial_1 g_{21} - \partial_2^2 g_{11} + \partial_1 \partial_2 g_{12} - \partial_1^2 g_{22}) - g_{ef}(\Gamma_{11}^e \Gamma_{22}^f - \Gamma_{12}^e \Gamma_{21}^f) \\ &= -\frac{1}{2}\partial_2^2 g_{11} - g_{11}(\Gamma_{11}^1 \Gamma_{22}^1 - \Gamma_{12}^1 \Gamma_{21}^1) - g_{22}(\Gamma_{11}^2 \Gamma_{22}^2 - \Gamma_{21}^2 \Gamma_{21}^2) \\ &= a^2 \sin^2 \theta.\end{aligned}$$

Thus the Gaussian curvature K of a spherical surface is given by

$$K = \frac{R_{1212}}{g} = \frac{a^2 \sin^2 \theta}{a^4 \sin^2 \theta} = \frac{1}{a^2}.$$

Instead of a spherical surface, we could instead consider the surface of a cylinder of radius a . The metric of the surface in cylindrical polar coordinates is

$$ds^2 = a^2 d\theta^2 + dz^2,$$

and it is obvious that this two-dimensional space is spatially flat because we can transform the metric into the form

$$ds^2 = dx^2 + dz^2$$

by the coordinate transformation $x = a\theta$. It therefore follows that the curvature of a cylindrical surface vanishes.

Exercises

- 7.1 From Poisson's equation $\nabla^2\Phi = 4\pi G\rho$ show that the gravitational potential outside a spherical object of mass M at a radial distance r from its centre is given by $\Phi(r) = -GM/r$. What is the form of $\Phi(r)$ inside a uniform spherical body?
- 7.2 A charged object held stationary in a laboratory on the surface of the Earth does not emit electromagnetic radiation. If the object is then dropped so that it is in free fall, it will begin to radiate. Reconcile these observations with the principle of equivalence. *Hint: Consider the spatial extent of the electric field of the charge.*
- 7.3 If X^μ is a local Cartesian coordinate system at some event P , show that so too is the coordinate system $X'^\mu = \Lambda^\mu{}_\nu X^\nu$, where $\Lambda^\mu{}_\nu$ defines a Lorentz transformation.
- 7.4 If two vectors $\mathbf{v}$ and $\mathbf{w}$ are Fermi–Walker-transported along some observer's world-line, show that their scalar product $\mathbf{v} \cdot \mathbf{w}$ is preserved at all points along the line.
- 7.5 Photons of frequency ν_E are emitted from the surface of the Sun and observed by an astronaut with fixed spatial coordinates at a large distance away. Obtain an expression for the frequency ν_O of the photons as measured by the astronaut. Hence estimate the observed redshift of the photon.
- 7.6 An experimenter A drops a pebble of rest mass m in a uniform gravitational field g . At a distance h below A , experimenter B converts the pebble (with no energy loss) into a photon of frequency ν_B . The photon passes by A , who observes it to have frequency ν_A . Use simple physical arguments to show that to a first approximation

$$\frac{\nu_B}{\nu_A} = 1 + \frac{gh}{c^2}.$$

Use this result to argue that for two stationary observers A and B in a weak gravitational field with potential Φ , the ratio of the rates at which their laboratory clocks run is $1 + \Delta\Phi/c^2$, where $\Delta\Phi$ is the potential difference between A and B .

- 7.7 A satellite is in circular polar orbit of radius r around the Earth (radius R , mass M). A standard clock C on the satellite is compared with an identical clock C_0 at the

south pole on Earth. Show that the ratio of the rate of the orbiting clock to that of the clock on Earth is approximately

$$1 + \frac{GM}{Rc^2} - \frac{3GM}{2rc^2}.$$

Note that the orbiting clock is faster only if $r > \frac{3}{2}R$, i.e. if $r - R > 3184$ km.

- 7.8 Consider the limit of a weak gravitational field in a coordinate system in which $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, with $|h_{\mu\nu}| \ll 1$, and $\partial_0 g_{\mu\nu} = 0$. Keeping only terms that are first order in v/c , show that the equation of motion for a slowly moving test particle takes the form

$$\frac{d^2 x^i}{dt^2} \approx -\frac{1}{2}c^2 \delta^{ij} \partial_j h_{00} + c \delta^{ik} (\partial_j h_{0k} - \partial_k h_{0j}) v^j.$$

Give a physical interpretation of the second term on the right-hand side.

- 7.9 Show that in a two-dimensional Riemannian manifold all the components of R_{abcd} are equal either to zero or to $\pm R_{1212}$.
- 7.10 Show that the line element $ds^2 = y^2 dx^2 + x^2 dy^2$ represents the Euclidean plane, but the line element $ds^2 = y dx^2 + x dy^2$ represents a curved two-dimensional manifold.
- 7.11 For a two-dimensional manifold with line element $ds^2 = dr^2 + f^2(r) d\theta^2$, show that the Gaussian curvature is given by $K = -f''/f$, where a prime denotes d/dr .
- 7.12 By calculating the components of the curvature tensor R^d_{abc} in each case, show that the line element

$$ds^2 = \frac{a^2}{a^2 - r^2} dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2$$

represents a curved three-dimensional manifold. Show that the manifold is flat in the limit $a \rightarrow 0$.

- 7.13 A spacetime has the metric

$$ds^2 = c^2 dt^2 - a^2(t)(dx^2 + dy^2 + dz^2).$$

Show that the only non-zero connection coefficients are

$$\Gamma^0_{11} = \Gamma^0_{22} = \Gamma^0_{33} = a\dot{a} \quad \text{and} \quad \Gamma^1_{10} = \Gamma^2_{20} = \Gamma^3_{30} = \dot{a}/a.$$

Deduce that particles may be at rest in such a spacetime and that for such particles the coordinate t is their proper time. Show further that the non-zero components of the Ricci tensor are

$$R_{00} = 3\ddot{a}/a \quad \text{and} \quad R_{11} = R_{22} = R_{33} = -a\ddot{a} - 2\dot{a}^2.$$

Hence show that the 00-component of the Einstein tensor is $G_{00} = -3\dot{a}^2/a^2$.

- 7.14 Show that the covariant components of the curvature tensor are given by

$$R_{abcd} = \frac{1}{2}(\partial_d \partial_a g_{bc} - \partial_d \partial_b g_{ac} + \partial_c \partial_b g_{ad} - \partial_c \partial_a g_{bd}) - g^{ef}(\Gamma_{eac} \Gamma_{fbd} - \Gamma_{ead} \Gamma_{fbc}),$$

and hence verify its symmetries. Show further that, for an N -dimensional manifold, the number of independent components is $N^2(N^2 - 1)/12$.

7.15 Show that for any two-dimensional manifold the covariant curvature tensor has the form

$$R_{abcd} = K(g_{ac}g_{bd} - g_{ad}g_{bc}),$$

where the scalar K may be a function of the coordinates. Why does this result not generalise to arbitrary manifolds of higher dimension?

7.16 If v^a are the contravariant components of a vector and T^{ab} are the contravariant components of a rank-2 tensor, prove the results

$$\begin{aligned}\nabla_c \nabla_b v^a - \nabla_b \nabla_c v^a &= -R^a_{dbc} v^d, \\ \nabla_d \nabla_c T^{ab} - \nabla_c \nabla_d T^{ab} &= -R^a_{ecd} T^{eb} - R^b_{ecd} T^{ae}.\end{aligned}$$

Can you guess the corresponding result for the mixed components T^{ab}_c of a rank-3 tensor?

7.17 Show that any Killing vector v^a , as defined in Exercise 4.11, satisfies the relations

$$\begin{aligned}\nabla_c \nabla_b v^a &= R^a_{bcd} v^d, \\ v^a \nabla_a R &= 0.\end{aligned}$$

7.18 Calculate explicit forms for the Ricci tensor R_{ab} and the Ricci scalar R in terms of the metric, the connection and its partial derivatives.

7.19 Prove that the Ricci tensor R_{ab} is symmetric.

7.20 A *conformal transformation*, such as that in Exercise 2.7, is not a change of coordinates but an actual change in the geometry of a manifold such that the metric tensor transforms as

$$\tilde{g}_{ab}(x) = \Omega^2(x) g_{ab}(x),$$

where $\Omega(x)$ is some non-vanishing scalar function of position. Show that, under such a transformation, the metric connection transforms as

$$\tilde{\Gamma}^a_{bc} = \Gamma^a_{bc} + \frac{1}{\Omega} (\delta^a_c \partial_b \Omega + \delta^a_b \partial_c \Omega - g_{bc} g^{ad} \partial_d \Omega).$$

Hence show that the curvature tensor, the Ricci tensor and the Ricci scalar transform respectively as

$$\begin{aligned}\tilde{R}^a_{bcd} &= R^a_{bcd} - 2 \left(\delta^a_{[c} \delta^e_{d]} \delta^f_b - g_{b[c} \delta^e_{d]} g^{af} \right) \frac{\nabla_e \nabla_f \Omega}{\Omega} \\ &\quad + 2 \left(2 \delta^a_{[c} \delta^e_{d]} \delta^f_b - 2 g_{b[c} \delta^e_{d]} g^{af} + g_{b[c} \delta^a_{d]} g^{ef} \right) \frac{(\nabla_e \Omega)(\nabla_f \Omega)}{\Omega^2}, \\ \tilde{R}_{bc} &= R_{bc} + \left[(N-2) \delta^f_b \delta^e_c + g_{bc} g^{ef} \right] \frac{\nabla_e \nabla_f \Omega}{\Omega} \\ &\quad - \left[2(N-2) \delta^f_b \delta^e_c - (N-3) g_{bc} g^{ef} \right] \frac{(\nabla_e \Omega)(\nabla_f \Omega)}{\Omega^2}, \\ \tilde{R} &= \frac{R}{\Omega^2} + 2(N-1) g^{ef} \frac{\nabla_e \nabla_f \Omega}{\Omega^3} + (N-1)(N-4) g^{ef} \frac{(\nabla_e \Omega)(\nabla_f \Omega)}{\Omega^4},\end{aligned}$$

where N is the dimension of the manifold.

- 7.21 Show that parallel transportation of a vector around the closed triangle ABC on the surface of a sphere, as shown in Figure 7.3, results in a vector that is orthogonal to its original direction.
- 7.22 On the surface of a sphere, show that, along the geodesic $\phi = \text{constant}$, the geodesic deviation vector ξ^i satisfies

$$\frac{D^2 \xi^\theta}{Ds^2} = 0, \quad \frac{D^2 \xi^\phi}{Ds^2} = -\xi^\phi \left(\frac{d\theta}{ds} \right)^2.$$

Choose a geodesic $\phi = \phi_0$ with path length $s = \theta$ measured from $\theta = 0$, and a neighbouring geodesic $\phi = \phi_0 + \delta\phi_0$, also with $s = \theta$, and define $\xi^i(\theta)$ as the vector between $s = \theta$ on one and $s = \theta$ on the other. Show that $\xi^i(\theta) = (0, \xi^\phi)$ for all θ . Show in addition that if $\xi^\phi = 0$ when $\theta = 0$ then

$$\xi^\phi \xi_\phi = l^2 \sin^2 \theta,$$

where l^2 is a constant, and that the two geodesics pass through $\theta = \pi$.

- 7.23 In Newtonian gravity, consider two nearby particles with trajectories $x^i(t)$ and $\bar{x}^i(t)$ ($i = 1, 2, 3$) respectively in Cartesian coordinates. Show that the components of the separation vector $\zeta^i = x^i - \bar{x}^i$ evolve as

$$\frac{d^2 \zeta^i}{dt^2} = - \left(\frac{\partial^2 \Phi}{\partial x^i \partial x^j} \right) \zeta^j,$$

where Φ is the Newtonian gravitational potential.

- 7.24 In the weak-field, Newtonian, limit of general relativity, we may choose coordinates such that $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, where $|h_{\mu\nu}| \ll 1$, and we assume that all particle velocities are small compared with c . By considering the equation of geodesic deviation, show that the general-relativistic tidal force reduces to the Newtonian limit given in Exercise 7.23.